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Sound, color and meaning in the metallurgy of Ancient West Mexico

Dorothy Hosler

Abstract

West Mexican metallurgy developed between AD 600 and the Spanish invasion. Laboratory evidence shows that metalworkers used a variety of metals and alloys, including copper-tin and copper-arsenic bronze, to craft both tools and status items. None the less, the primary technical focus was on two non-utilitarian properties of metal, its sound in bells and its golden and silvery colors in bells and sheet metal ornaments. Ethnohistoric and linguistic evidence indicates that the sounds of bells offered protection in war and structured rituals that celebrated fertility and regeneration. Metallic colors, especially of gold and silver, were associated with the solar and lunar deities and a shimmering sacred paradise. In focusing on metallic sounds and metallic colors, this metallurgical technology in large measure constituted a system of visual and aural symbols that created and recreated the experience of the sacred.

Keywords

Metallurgy; Mesoamerica; West Mexico; metallic sound; metallic color.

West Mexican metallurgy developed between AD 600 and the Spanish invasion in 1521 in a region I have designated the West Mexican metalworking zone (Hosler 1994) (Fig. 1). Metalworkers in certain sectors of this zone eventually used a host of metals and alloys, including the bronzes – copper-arsenic and copper-tin – a ternary copper-arsenic-tin alloy, alloys of copper and silver, and others to craft a variety of object classes, from tools to élite-status items. Some of these were cold worked from an initial cast blank, others were worked hot, and still others were lost-wax cast. Laboratory analytical studies demonstrate these metalworkers' technical versatility (see Hosler 1986, 1988a, 1988b; Hosler, Lechtman and Holm 1990; Hosler and Stresser-Péan 1992). The studies further reveal that the artisans were most consistently interested in elaborating two physical properties of metal, its sound and its color. Lost-wax-cast bells and, much less frequently, other sorts of metallic rattling artifacts comprise the sounding instruments. The focus on color, especially the colors of gold and silver, is apparent in bells and in other status items such as large ornamental tweezers, rings and ornamental shields fashioned from copper alloys by hot or cold work. When made from bronze, these artifacts contain the alloying element, tin

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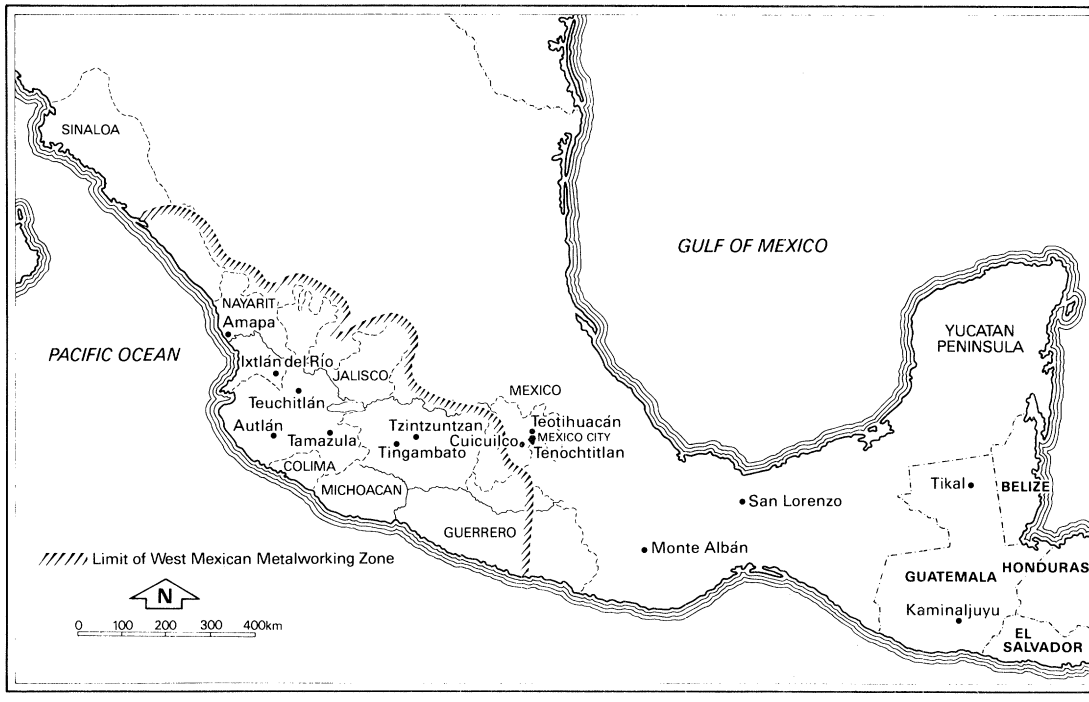

Figure 1 Map of Mesoamerica showing archaeological sites, the state boundaries in western Mexico and the limits of the West Mexican metal-working zone. Courtesy of MIT Press.

or arsenic, in concentrations between 5 and 23 weight per cent, levels which laboratory studies (Hosler 1986) have shown are far higher than necessary solely to optimize artifact design and mechanical function. These levels of tin or arsenic dramatically alter the color of copper metal, however, from red to increasingly golden or silvery hues.

Artisans also used copper alloys, especially the bronzes, for utilitarian objects, such as axes, needles and awls. These contain tin and arsenic in concentrations between approximately 2 and 5 weight per cent. At these levels, alloy concentration is sufficient to confer the increased strength required to improve performance and design without appreciably affecting color.

Technical developments of this sort are especially compelling when we consider that the geological, metallurgical and chemical analytical data (Hosler 1986, 1988a, 1988c, 1994; Hosler and Stresser-Péan 1992) indicate metalworkers in this zone had a range of other options available to them. Surveys of regional geology show that the area contains abundant deposits of native metals and ore minerals. The data also show that these artisans fully understood and managed alloy properties, especially of the bronzes, and could have employed metal extensively or perhaps exclusively for tools, weapons, armor and other utilitarian objectives. Yet needles, punches, fishhooks and other tools are small in number relative to the numbers of status items in metal (see Hosler 1986, 1994). The most original technical experiments of these smiths and the bulk of their technical energies were devoted

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to developing sound and specific colors in objects worn by élites and nobles and used in ritual.

Here I explore the meaning of bell sounds and the colors of gold and silver in an effort to explain why these sounds and these colors played such key roles in shaping this prehistoric metallurgy. The importance of metallic colors, especially of gold and silver, in myth, shamanistic performance, cosmological schema and political ideology has been demonstrated for the central Andes and Colombia (for example, see Lechtman 1979, 1993; Reichel-Dolmatoff 1988). The West Mexican experiment is unique in that artisans achieved these culturally required golden and silvery colors through the unusual technical expedient of using the high-arsenic and high-tin bronzes. Although the mechanical properties of these alloys were required by the design characteristics of the objects, their colors were a matter of choice, accomplished by adding the alloying element in high concentrations. By generating metallic sound and these metallic colors, this ancient technology served primarily as a visual and auditory system that symbolically defined élite and sacred spheres of activity.

An overview of West Mexican metallurgy

Metallurgy emerged late in ancient Mesoamerica, appearing after AD 600 in the mineral rich western states of Michoacan, Jalisco, Colima, Nayarit and Guerrero. The technology occurred after state-level societies had been in place in adjacent Mesoamerican regions for at least several hundred years. This western area had been occupied by smaller-scale societies; larger-scale social entities did not take shape until the Tarascan state coalesced in highland Michoacan around AD 1250.

West Mexican metallurgy developed through two technological periods, each of which had its roots in the metallurgies of Central and South America (Hosler 1988a, 1988b). Elements of both these two more southerly metallurgies were introduced via a maritime route to West Mexico. During Period 1 (AD 600–1200/1300), West Mexican smiths were particularly engaged in elaborating metals' resonant qualities, casting bells in copper in a variety of shapes and sizes using the lost-wax method (Plates 1 and 2). They also made various tools and ornaments by cold working them to shape (Plates 3 and 4), but the relative numbers of these items in collections and excavated assemblages is far lower than that of bells, which comprised the focus of these artisans' technical activities (Hosler 1994). Lost-wax casting, particularly of certain bell types, had been introduced from lower Central America and Colombia. Cold working, especially of several varieties of small hand tools (i.e. needles, tweezers, etc.), was introduced from southern Ecuador and northern Peru. During Period 1, West Mexican smiths primarily used copper, although some evidence indicates that they also used a low-arsenic copper-arsenic alloy, and occasionally made objects from silver and from gold.

During Period 2 (AD 1200/1300 to the Spanish invasion), these smiths' technical repertoire greatly expanded. Michoacan, the seat of the Tarascan empire, seems to have been the locus of Period 2 technical experiments, although the artifacts also appear at sites in adjacent regions of Guerrero, Jalisco and the State of Mexico (see Fig. 1) (Hosler 1994). Metalworkers used alloys (copper-tin, copper-arsenic, copper-arsenic-tin, copper-silver

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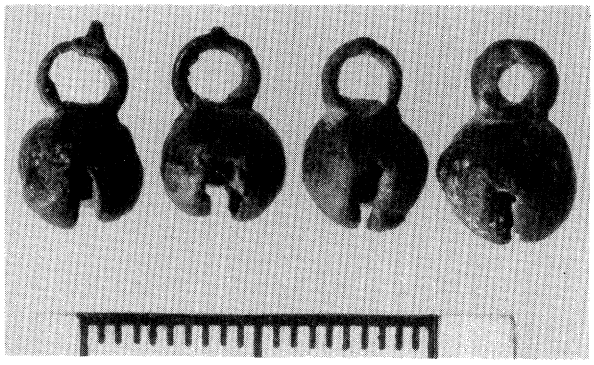


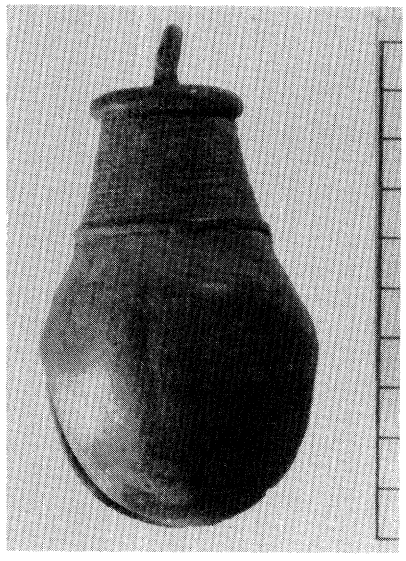
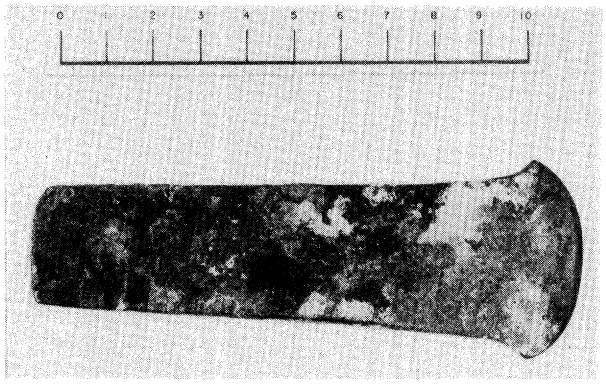


Plate 1 Period 1 copper bells from West Mexico cast using the lost-wax method. From the collections of the Regional Museum of Guadalajara, Guadalajara, Jalisco Mexico.
Plate 2 (right) Period 1 copper bell from West Mexico cast using the lost-wax method. From the collections of the Regional Museum of Guadalajara, Guadalajara, Jalisco Mexico.



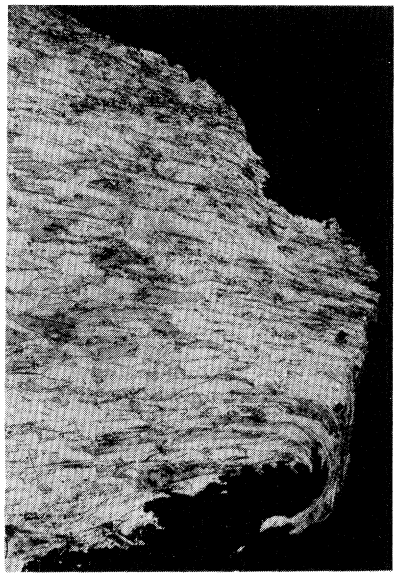


Plate 3 Period 1 copper axe from West Mexico. From the collections of the Regional Museum of Guadalajara, Guadalajara, Jalisco Mexico.
Plate 4 (right) Longitudinal sample from blade tip of a Period 1 copper axe. Elongated and compressed grains reveal heavy cold work and the extreme tip shows deformation from use (mag. 50×).

and others) to redesign some of the objects they had previously made in copper, taking advantage of the alloys' strength, toughness, fluidity and color properties. Artisans added the alloying elements, tin and arsenic, to copper in low concentrations for tools such as axes which they were able to make thinner and harder. They took advantage of the same alloy properties for status objects, using tin bronze and arsenic bronze to cast thin-walled bells with intricate designs that required improved strength and solidification characteristics, for worked objects such as thin-banded rings and for large tweezers with convex shell-design blades (Hosler 1988a). As mentioned, however, the concentration of the alloying elements in such status objects was consistently higher than necessary to achieve

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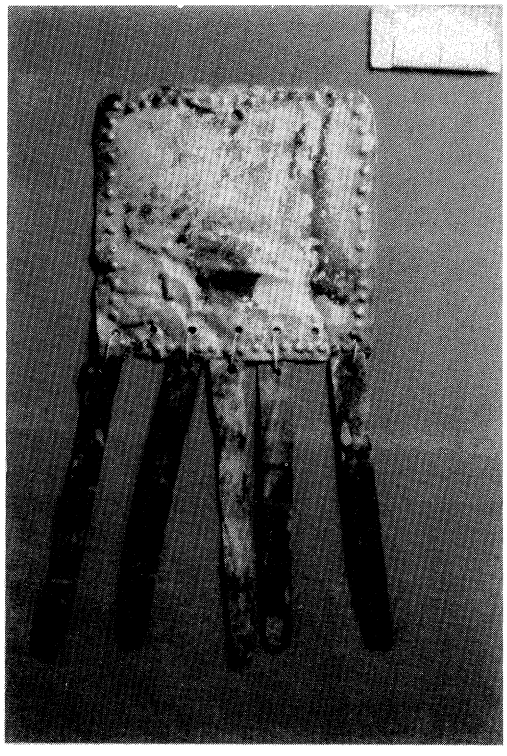

Plate 5 Period 2 silver-copper sheet metal ornament. From the collections of the Regional Museum of Guadalajara, Guadalajara, Jalisco Mexico.

these designs; what it did affect was color. For silvery looking castings smiths used arsenic bronze, but they preferred the tough copper-silver alloys for sheet metal objects: discs, pendants and other ornaments. In fact, copper-silver alloy hammered into sheet-metal ornaments (Plate 5) was so common in Michoacan that the Spaniards referred to the alloy as the 'metal of Michoacan' (Warren 1985).

Certain elements of Period 2 West Mexican metallurgy also were introduced from the metallurgies of Central and South America. Knowledge of the copper-arsenic and copper-silver alloy systems and of certain artifact designs derives from the metallurgies of southern Ecuador and northern Peru. The copper-tin bronze alloy, however, was introduced from the southern highlands of Peru and Bolivia (Hosler 1988b) through the same maritime network that transmitted knowledge of this technology during Period 1.

At no time during these developments did West Mexican smiths import either artifacts or raw materials from the south. What they imported was knowledge of fabrication regimes and a few prototype artifacts that were copied locally. They also reproduced the pan-American indigenous interest in metallic color, especially of gold and silver, although they invented their own methods for achieving these colors. The laboratory investigations described here have identified not only the characteristics of the technology, but its cultural bias. The technology itself provides the point of departure for this inquiry into meaning.

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The meanings of bell sounds and of the colors of gold and silver

Sources

How do we begin to understand why such a large fraction of this metallurgy was devoted to elaborating these properties? What specifically did golden and silvery metallic colors and the sounds of bells mean to these ancient peoples? Sixteenth-century documents are particularly helpful in establishing the social contexts in which metallic sound and silvery and golden metallic colors played a part, at least during Period 2. Another, complementary way of approaching the problem is through indigenous terms related to metal and sounding instruments, in these documents and in Tarascan (Purepecha) and Nahuatl, two languages spoken in the zone. Tarascan was spoken by the peoples in highland Michoacan and adjacent areas; Nahuatl was the language of the Aztec or Mexica. Nahuatl speakers occupied some areas of Guerrero lying within the metalworking zone and areas contiguous with it. Nahuatl was also spoken in some areas of Michoacan (Fig. 1), especially along the Tarascan border in certain multiethnic communities. Among sixteenth-century Nahuatl documents, Bernardo de Sahagún's encyclopedic *Florentine Codex* (1950–82) is especially helpful. Documentary sources are also available from the Tarascan region and other areas of the metalworking zone. The most useful of these is the *Relación de Michoacán* (Tudela 1977), an anonymous illustrated sixteenth-century Spanish manuscript treating preconquest Tarascan and very early colonial history.

The colors of gold and silver

Golden and silvery metallic colors achieved through alloying, or, where possible, with the pure metals were one primary technical objective of the smiths of this zone during Period 2. Why were these peoples so interested in the colors of gold and silver? The documentary evidence suggests that Tarascan peoples considered the golden and silvery objects divine and that they associated gold with the sun and silver with the moon. The *Relación de Michoacán* recounts that Hiripan, one of three brothers responsible for preconquest Tarascan expansion, wanted to recover gold, silver and other items from conquered villages.

> 'What shall we do brothers? The people of the villages run away and carry off the plumes and jewels which made them nobles in the villages we have conquered. Go get them so the gods [the gold and silver objects] come back to their villages.' Seeing that yellow gold and white silver, Hiripan said, 'Look brothers, this yellow [metal] must be manure which the sun casts off, and that white metal must be manure cast off by the moon'.
>
> (Tudela 1977: 152)

These two metals, gold and silver, divine emissions and divine property, also validated the authority of the nobility. The citation suggests that nobility defined itself through the possession of these plumes and jewels. Linguistic data reinforce the association of the sun with gold; the root *tiripeti* means gold in Tarascan, and *Tiripeti* was also the name given to gods that were individual manifestations of the sun (Brand 1951). In fact, Tarascan state treasurers stored gold diadems and discs in chests said to honor the sun, and discs of silver

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in honor of Xaratanga, the goddess of the moon (Tudela 1977: 257). Tarascan kings were interred with gold shields at their back and golden bells on their ankles (Tudela 1977: 219); in battle they were protected by a shield of silver (Tudela 1977: 192). Some of these objects were made from the pure metals. Others were apparently made from copper-silver and, probably, gold-silver alloys based on sixteenth-century accounts of the variable quality of the gold and silver objects which the Spaniards demanded and received from the Tarascan ruler.

Golden and silvery metallic colors also were associated with solar and lunar deities by the Aztec or Mexica in central Mexico, and by extension by Nahuatl-speaking peoples in surrounding territories. In Nahuatl, the word for gold, *cuztic teocuitlatl*, literally means 'yellow divine excretions' and is taken to mean excretions of a solar deity (Campbell 1985). According to the *Florentine Codex*, that deity was Tonatiuh. The name Tonatiuh comes from the verbal root *tona* which means to shimmer, to shine or to give off heat. When the *Florentine Codex* describes the visual effect of gold, it uses *tona*, translating it to mean 'to give off rays' (Sahagún 1950–82 book 11: 234). *Tona* also means for the sun to shine; thus gold shines, shimmers and gives off rays, or heat, like the sun or like *Tonatiuh*. Metal conducts heat and it is also highly reflective and here both qualities are explicitly recognized. The origins of gold are also described in the *Florentine Codex*: the name 'derives from [the fact that] sometimes, in some places, there appears in the dawn something like a little bit of diarrhea. . . . It was very yellow, very wonderful, resting like an ember, like molten gold' (Sahagún 1950–82 book 11: 233). The Nahuatl word for silver, *iztac teocuitlatl*, signifies 'white divine excretions' (Campbell 1985), and could mean excretions of the moon. The terms make clear that these metals were divine substances, produced, smelted or emitted by the deities (Hosler 1994).

Golden and silvery metal objects such as discs, crowns, large tweezers and pendants made from high-tin bronze, copper-silver alloys and sometimes pure gold or silver metals thus announced status. Status resided in supernatural affiliations. The quality of brilliance and reflectiveness of these objects characterizes the Mexica (Aztec) paradise, termed a 'cult of brilliance' (Burkhart 1992; Hill 1987). In Nahuatl devotional literature this sacred domain is conceived as a shimmering garden. Burkhart describes the garden.

> In this symbolic garden, one came into direct contact with the creative, life-giving forces of the universe and with the timeless world of deities and ancestors. The garden is a shimmering place filled with divine fire; the light of the sun reflects from the petals of flowers and the iridescent feathers of birds; human being – the souls of the dead or the ritually transformed living – are themselves flowers, birds, and shimmering gems. . . . This garden is not a place of reward for the righteous, existing on some transcendent plane of reality separate from the material world. It is a metaphor for life on earth, a metaphor that ritual transforms into reality by asserting that, in fact, this is the way the world is.
>
> (Burkhart 1992: 89)

Nahuatl devotional texts present the entities that fill this world as feathers, bells, flutes and flowers. Metallic colors reflecting off these iridescent entities helped create the sacred garden. These same texts frequently describe glowing and shining golden blossoms, golden petals, leaves, birds, feathers and raindrops. The examples Burkhart discusses are

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predominantly drawn from two Nahuatl texts, *Psalmodia Christiana*, a hymnbook composed by Fray Bernardino de Sahagún in Nahuatl to christianize the native peoples of Mexico (Sahagún 1993) and *Cantares Mexicanos* (Bierhorst 1985), a book of Nahuatl poetry composed between 1550 and 1580 dealing with the conquest and its aftermath.

We do not yet know whether a similar sacred domain also existed in Tarascan conceptions of the divine, although it seems likely given the vast quantities of golden and silvery sheet-metal ornaments Tarascan metalworkers made and the sacred qualities ascribed to golden and silvery colors. We suspect that the concept of a brilliant shimmering paradise was held elsewhere in Mesoamerica, certainly by other Uto-Aztecan speakers (Hosler 1994), and possibly among the Maya (Burkhart personal communication 1993). Some evidence also indicates that these same concepts were present in Andean South America. The implications of an Andean South American locus for these ideas is especially intriguing in view of the Andean peoples' widespread and very early focus on producing reflective golden and silvery metallic sheet – and the fact that aspects of that same sheet-metal-oriented ritual technology were introduced from the Andean area to western Mexico.

Achieving golden and silvery metallic colors, a primary technical objective of West Mexican smiths, was a means of visually creating and recreating the sacred, the essence of these deities. The sacred was embodied in a constellation of golden and silvery looking objects such as tweezers, bells, discs, rings, plaques and ornamental shields. The objects, which were made from the sacred metals of gold and silver, or their alloys with copper, were possessed by these divinities and the kings and nobles who represented them.

Bell sounds

Bell sounds and the sounds of composite bell instruments played a significant role in several sacred contexts. One is in ritual celebrating human and agricultural fertility and regeneration. Another is in warfare, where bell sounds could protect. The third is in the sacred paradise, which was created through song and sound. One of those sounds, the sound of bells, is associated with shimmering, colorful, singing birds and with human voices that represent deities and their human transformations.

The power accorded to bell sounds also determined the course of West Mexican metallurgy. Our best information comes from central Mexican descriptions of ritual in which these instruments are played. Rites for Tlaloc, Xipe Totec and Quetzalcoatl, three central Mexican deities, invariably require bells or instruments containing them. Although not all central Mexican deities were worshipped in the Tarascan region, aspects of these three are represented by deities in the Tarascan pantheon, and Tarascan ideology shares basic principles with other Mesoamerican societies including auto-sacrifice, flaying, the ball game and heart sacrifice (Pollard 1993). Ritual surrounding these events was broadly similar, as were ritual accouterments, including bells and other percussion instruments. We also know that the musical cultures of Central Mexico (Aztec), Michoacan (Tarascan) and Oaxaca were fundamentally similar (Stanford 1966), so that the instruments and the meanings ascribed to their sounds also resemble one another.

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Fertility and regeneration

Rituals for Tlaloc, Xipe Totec and Quetzalcoatl representing aspects of fertility, life and regeneration are frequently created through bell sounds and the sounds of other composite percussion instruments containing bells. Rain, water, storms, thunder, lightning, rattlesnakes and new vegetation repeatedly appear as symbols of fecundity and new life and are associated with these deities. Bells and other rattling instruments played in these rituals replicated the sounds of thunder, rain and the rattle of the rattlesnake.

In addition to bells, the percussion (rattling) instruments most frequently illustrated or described from Central Mexico and surrounding territories are the hand-held rattle, the rattlestick and the mist rattleboard. Terms for some of these instruments, at least in Nahuatl, connote fertility. The Aztec rattlestick or *chicauaztli* (Stevenson 1968), a long, pointed stick containing rattles or bells, was widely employed in agricultural rites. *Chicaua*, the verbal root of the noun *chicauaztli*, means to strengthen and fortify (Karttunen 1992: 46). Seler argues that the 'word and symbol obviously refer to the strengthening of the reproductive function, to fertilizing' (in Sahagún 1950–82 book 1: 40, note 118). The *Codex Borgia* illustrates a rattlestick above the first human pair, who appear to be covered with a common quilt, explicitly linking the instrument with human sexuality (Neumann 1974). Neumann thinks that the *chicauaztli* serves as a symbol for sexual union. The term for mist rattleboard, *ayauhchicauaztli*, also contains *chicaua*, the same verbal root. Seler describes the instrument as a board with holes. Bells were strung in them, and the bells sounded as the person carrying it walked (in Sahagún 1950–82 book 1: 17, note 53). Seler says that the sounds reproduced the thunder of Tlaloc, who, according to Pasztory (1974), is a rain deity in one aspect and a deity of the earth, caves and the underworld in another.

Rites for Tlaloc, associated with lightning, rattlesnakes and the underworld, are accompanied by instruments containing bells or rattles to attract rain and thunder. The *Florentine Codex* describes the fire priest in Tlaloc's temple when the rains break out. The images graphically link snakes, their rattling sounds and rain.

> And when the rain broke out, then he forthwith arose; he seized his incense ladle. . . . The incense ladle rattled. It was in the form of a serpent. And the serpent's head also rattled. . . . Then he offered incense; to the four directions he raised [the incense ladle]. Much did it rattle; [the incense] spilled out. . . . Thus he attended to the matter; thus he called upon the Tlalocs; thus he prayed for rain.
>
> (Sahagún 1950–82 book 2: 151)

In rites honoring Tlaloc during *Etzalqualiztli*, the sixth month of the Aztec ritual calendar which marked the end of the dry season, priests carried and shook the *ayauhchicauaztli*, the mist rattleboard.

> He went bearing on his shoulders the mist rattleboard, also called the sorcerer's staff. It was wide, very wide, excessively wide; and it was long, very long. And it rattled; he went along rattling it.
>
> (Sahagún 1950–82 book 2: 81)

In describing a procession to the temple of Tlaloc, the priest

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> Put on his mist jacket, his rain mask or his Tlaloc mask.... Thus they went to the Temple of Tlaloc.... And when this was done, then he scattered *yauhtli* [an herb]. And when he had scattered it, then they gave him the mist rattleboard. He rattled, he shook it; he raised it in dedication [to the god].
>
> (Sahagún 1950–82 book 2: 87)

The final verse of the song of Tlaloc makes clear that rattle sounds bring rain.

> To all places go
> To all places reach
> To Poyauhtlan
> With mist-bringing rattles [*ayauhchicauztica*]
> To Tlálocan taken.
>
> (Sahagún 1950–82 book 2: 225)

Bells and bell sounds are also associated with Quetzalcoatl (the feathered serpent). In one of Quetzalcoatl's myriad aspects of wind and storm deity, he represents fertility, wind or breath, and life (Brundage 1982; Hunt 1977). Quetzalcoatl becomes the wind storm that occurs just before the rainy season begins. Sahagún calls him the 'roadsweeper' of the rain gods. According to Townsend (1992: 114) the best portrayal of Quetzalcoatl is an Aztec sculpture depicting a coiled serpent rising from the earth, with a Tlaloc mask on the bottom. Quetzalcoatl's twin, Xolotl, appears as a deified form of lightning (Hunt 1977: 126). As snake, Quetzalcoatl is often depicted with prominent rattles; the rattles are sometimes shown as bells (Emmerich 1965: Fig. 178; I. Kelly personal communication 1981). According to Sahagún, Quetzalcoatl in human form wears ankle bells. Quetzalcoatl occasionally is depicted holding the *chicauaztli*. Mayahuel, the goddess of *pulque* with 400 breasts whom Brundage (1982: 96) calls an avatar of Quetzalcoatl, holds the *chicauaztli* as well. Tezcatzoncatl, one of the rabbit gods, is also illustrated with ankle bells. Hunt thinks that these rabbit gods and gods of *pulque* also are linked to fertility (Hunt 1977: 86–7).

Xipe Totec, patron of the metalsmiths and god of new vegetation, is another deity invariably represented with percussion and rattling instruments. The *chicauaztli* is Xipe's insignia. Xipe's origins are confusing, but one possibility is that he came from the Yope-Tlapanec area of Guerrero, a zone within the metalworking area. Xipe is the principal deity of the Aztec second month, Tlacaxipeualiztli (the flaying of people), when sacrificial victims were flayed. Flaying ceremonies are related to agriculture, fertility and prosperity (Broda 1970). The objects associated with Itztli, a deity related to Xipe, make this obvious: he is depicted grasping a digging or planting stick, wearing a flayed skin and holding a basket of corn (Heyden 1986: 40). Xipe's song, collected by Sahagún, is an 'invocation for rain' and focuses on the growth of the maize plant (Broda 1970: 259). It is therefore not surprising that mist rattleboards were sounded during Tlacaxipeualiztli, the ceremonies dedicated to Xipe.

> And when it was this [time], the rattleboards were sown there at Yopico. And only they, the old men of the *calpulli* at Yopico, sat singing, sat rattling their rattleboards until the day was done.
>
> (Sahagún 1950–82 book 2: 57)

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The phrase 'the rattleboards are sown' refers to a dance with bells that commoners danced (Broda 1970: 229), but 'sowing' rattleboards also links Xipe to rain (Hosler 1994).

In the Tarascan realm, agricultural and rain-making rites must also have been accompanied by bell sounds as these peoples shared common musical cultures and ideologies with the Aztec. Tarascan musicians and religious functionaries used two other instruments as well: hand-held rattles and some other sort of rattle. Hand-held rattles, shaken by dancers, are illustrated in the *Relación de Michoacán* (Tudela 1977: 160), and another rattle is also described among metal objects exported from Michoacan by the Spaniards.

> Further, I received in native rattles of the same metal [copper-silver alloys] 4 arrobas and 33 pounds. These rattles are put together like the breast leather of a saddle with heavy cords.
>
> (Warren 1985: 119)

Although Aztec deities and divinities in the Tarascan pantheon do not directly correspond to one another, certain Tarascan deities are associated with rain and agricultural and human fertility; the bells that were so pervasive in the region must have played a part in rites dedicated to them. The deity controlling rain and fertility is Cuerauaperi, who also controlled birth and death; she was the mother of all gods and venerated throughout the Tarascan realm. Pollard (1993) thinks that her worship was widespread by the late Preclassic or Formative period. Flaying rites, analogous to those carried out for Xipe, were dedicated to her, and she was also associated with snake imagery. Curicaueri, the sky god, was the sun's messenger, a warrior, the god of the hunt, and the patron god of the Tarascan royal dynasty. Xaratanga, the moon diety, was the wife of Curicaueri and daughter of the earth goddess; she is depicted as snake, half moon coyote and vulture. Xaratanga was associated with childbirth and fertility. Although we know something about ceremonies held for them we simply do not have the detail that is available for the Aztec (Hosler 1994).

The linguistic evidence for Tarascan is not as rich as for Nahuatl. None the less, certain terms, specifically for copper and rattlesnakes, are to some extent related and suggest that sounds made by coppery objects, most likely bells, reproduced the sound of the rattlesnake. *Shari* is a root for reddish objects, including rattlesnakes, and *charha* is one for red earth, including copper. A stronger relationship exists between the terms meaning metal and metal bell, suggesting that the purpose of metal was to make bells.

> *tiamu*: Hierro, campana, etc. [iron, bell, etc.]
> *tiamu charapeti*: Hierro, cobre. [iron, copper.]
>
> (Anonymous 1991 vol. 2: 597)

The sounds of bells as protection

Bell and other rattling sounds not only attract rain, they also protect in warfare. The Aztec god of war Huitzilopochli wore ankle bells. In a mythic battle with his 400 brothers, Sahagún (1950–82 book 3: 3) tells us that they arrayed themselves as if for war, binding little bells to the calves of their legs. None the less Huitzilopochli quickly dispatched them. The event is commemorated in myth, translated by León-Portilla and discussed by Matos M. (1987).

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In vain they tried to do something against him,
In vain they turned and faced him,
To the sounds of their bells,
And they slapped their shields.
They could do nothing,
They could achieve nothing,
They could defend themselves with nothing.

Huitzilopochli then arrays himself with their regalia which may explain why he is shown wearing bells.

He took from them their finery, their adornments,
Their destiny, put them on, appropriated them,
Incorporated them into his destiny,
Made of them his own insignia.

(Matos M. 1987: 53–4)

Bell and rattle sounds are also associated with warfare in the Tarascan region. For example, the *Relación de Michoacán* (Tudela 1977: 190) illustrates the chief warrior wearing an ankle band with bells attached as he and his men attack a village. It also reports that when Curatame, the son of the Tarascan leader Tariacuri, dressed for war, he hung from his temples many snake rattles that dangled by their tails (Tudela 1977: 132).

The sounds of bells: the sacred garden

Another arena in which bell sounds shaped experience of the sacred is in the Aztec divine garden that Burkhart describes. The garden, while filled with luminous beings and entities, comes to be through song or sound. Bell sounds, bird songs and the human voice singing represented the agency of creation. The quotation below illustrates the metaphorical associations among them. The key verb, *icauaca*, means also to warble, murmur or clamor (Campbell 1985: 114).

The spirit swans are echoing me as I sing, shrilling [warbling] like bells from the Place of Good Song.

(Bierhorst 1985: 141)

Birds, songs, flowers, trees and golden colors are all associated with bell sounds in the Aztec paradise.

Hear it! He's shrilling, warbling on the branches of the flower tree. He's shaking! It's the golden flower-bell, the rattle hummingbird, the swan, Lord Monencauhtzin. Like a gorgeous troupial fan he spreads his wings and soars beside the flower drum.

(Bierhorst 1985: 165–7)

Speech is also associated with birds, song and bell sounds.

May your speech resound! May there be chattering, may your songs resonate like bells!

(Burkhart 1992: 95)

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Other lexical evidence also shows that rattles, bells and rattling sounds are linked to song and related to speech. A single word means a clear sound, like a bell, and metallic sounds.

> *tzilictic*: Something which has a clear sound, like a bell or something similar.
> *tzilini*: For metal to sound or ring.
> *tzilinia*: To ring a bell or something similar.
>
> (Campbell 1985: 380)

In fact, the root *nahuatl*, meaning clear sound or order, also appears in compound nominals, one of which (*nauatillalia*) means town statutes; used in derived verb forms such as *nauati* (to speak loudly, or for a bell to have a good sound, or something similar), it can mean 'something that sounds good like a bell or a man who speaks well' (Campbell 1985: 201). These definitions link good bell sounds, order and laws or statutes; they suggest that sound transforms, it creates (Hosler 1994).

Sound and color

The idea that metallic sound can create metallic colors appears in these texts; it is through song and sound that the sacred shimmering garden comes into being, as the following stanza from *Cantares Mexicanos* shows.

> As colors I devise them. I strew them as flowers in the Place of Good Song. As jewel mats, shot with jade and emerald sunray, the Green Place flower songs are radiating green. A flower incense, flaming all around, spreads sky aroma, filled with sunshot mist, as I, the singer, in this gentle rain of flowers sing before the Ever Present, the Ever Near.
>
> (Bierhorst 1985: 141)

The same idea appears in other metaphors. These also show, however, that sound and color can come into being through precisely the same process, and that process can be a metallurgical one.

> I drill my songs as though they were jades. I smelt them as gold. I mount these songs of mine as though they were jades.
>
> (Bierhorst 1985: 207)

Smelting, that is winning metal from its ore through heat, gives birth to sound (and song), and it also gives birth to the golden metallic colors. Sound and golden metallic colors come into being in the same way. The material result is metallic sound-and-color objects: the high-tin and high-arsenic bronze bells that constituted one of the unusual technical achievements of Period 2 smiths.

The associations of metallic sound and metallic colors also appear in relation to Xipe, god of the metalsmiths. Xipe's name may derive from Nahuatl terms related to foreskin, to scrape or peel, to the head of the penis (Campbell 1985) and to layer. His insignia is the *chicauaztli*. Heyden (1986: 38) and others have noted that the rattlestick, which initiates thunder to attract rain, is an analog to the phallus. The rattlestick is sometimes represented as a serpent. Xipe is the god who loses (or peels) the skin, or puts on a layer of skin. Xipe can also be the rattlesnake that loses its skin but grows a new one. The god represents the

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earth that loses its skin after the harvest, and gains a new one with new vegetation, with the new rains. The new rains are brought about by the rattling of the rattlestick. Xipe is also the god of metalworkers. Metals are something that comes from the earth and the gods. They are divine emissions, powerful substances that are lost, shed and excreted but that may also provide a new layer, such as the silvery surface layer of copper-silver alloys.

The symbolic in West Mexican metallurgy

I have explored the question of the meaning, in ancient West Mexico, of the two elements that shaped the development of this prehistoric technology. Both pertain to the domain of religious ideology, and they are related to one another. Sound, here metallic sound, created color, particularly metallic color, at least in this partial interpretation of Nahuatl devotional poetry. In considering the varieties and numbers of metal artifacts thus far present in the West Mexican corpus, only a small component (certain tools) belonged to strictly utilitarian spheres. The vast majority of the items fashioned during Period 2 were either sounding objects, golden or silvery colored objects, or both.

I have argued that the sounds of rattles and bells promote human and agricultural fertility, they create order and they protect. They also create color, a glittering sacred paradise full of lustrous beings. The importance of bells is not only apparent in their variety and abundance, and the ways they are used in specific rites, but is encoded in language. The material, metal, and a kind of object made from it, bells, were inextricably associated. Metal and bells were cultural synonyms. The golden and silvery colors of some bells amplify their sacred and powerful qualities. The other golden and silvery objects that appear in the West Mexican repertoire may represent and create the shimmering sacred garden, through their reflective qualities and because gold and silver are specifically identified with the solar and lunar deities.

The West Mexican case strikingly demonstrates the ways in which symbolic and ideological factors can shape technologies. The relative weight of ideological as opposed to other factors varies with the particulars of the social and environmental circumstances. I touch on this problem for the West Mexican case elsewhere (Hosler 1994). Unfortunately we do not yet have the cross-cultural archaeological and technical data necessary to understand in a more general way how these variables interact; under what circumstances certain factors take precedence over others. What is clear from this discussion is that this ancient metallurgy emphasized certain physical properties, sound and color, which expressed fundamental religious beliefs, and that those beliefs were embedded in, and perpetuated through the technology and its products.

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